

Assignment-03

SET-03

CSA5113 - Cryptography

Karthheeka Repalle

192372289

CSE-A1

as/100

A system uses HMAC with SHA-256 to authenticate messages between a client and a server.

a) what is the purpose of HMAC over plain hash functions?

b) Given

- Message = "HELLO"
- Key = "SECRET"
- Hash Used = SHA-256

Explain how HMAC will be calculated.

c) why is it not advisable to use MD5 in HMAC for security-critical application

d) How does a MAC differ from a digital signature in terms of authentication and non repudiation.

a) HMAC "stands" for Hash-based message Authentication Code. It is a cryptographic mechanism used to provide message integrity and message authentication between two parties that share a secret key.

HMAC solves this by combining a key with secret and cryptographic hash function.

Only Sender & Receiver who know the shared secret key can generate the correct HMAC.

message + Secret Key



HMAC SHA-256



Authentication Code.

HMAC vs Plain Hash :-

Plain SHA-256 :-

message



SHA-256



Hash value

• Anyone can calculate the Hash

HMAC SHA-256 :-

message + Secret Key



HMAC-SHA-256



Authentication code

• only some one with Secret key, can normally generate a valid HMAC

∴ HMAC provides an important security advantage over a plain hash when authentication is required.

Given

message = "HELLO"

key = SECRET

Hash = SHA - 256

Formulae: $HMAC(k, m) =$

$$H((k' \text{ XOR } \text{Opad}) \parallel H(k' \text{ XOR } \text{ipad} \parallel m))$$

where,

k = secret key, k' = key adjusted to hash block

m = message, H = SHA - 256, ipad = inner padding -

STEPS :-

(i) KEY = SECRET

key is converted into byte representation

$k' = \text{Padded SECRET}$

(ii) Inner padding key.

$\text{ipad} = 0x36$ repeated for block size

Inner key = $k' \text{ XOR } \text{ipad}$

(iii) Add message

$(k' \text{ XOR } \text{ipad}) \parallel \text{HELLO}$

(iv) Calculate Inner SHA - 256 hash

SHA - 256 $((k' \text{ XOR } \text{ipad}) \parallel \text{"HELLO"})$

(v) Generate outer padded key.

Opad = 0x5C repeated for the block size

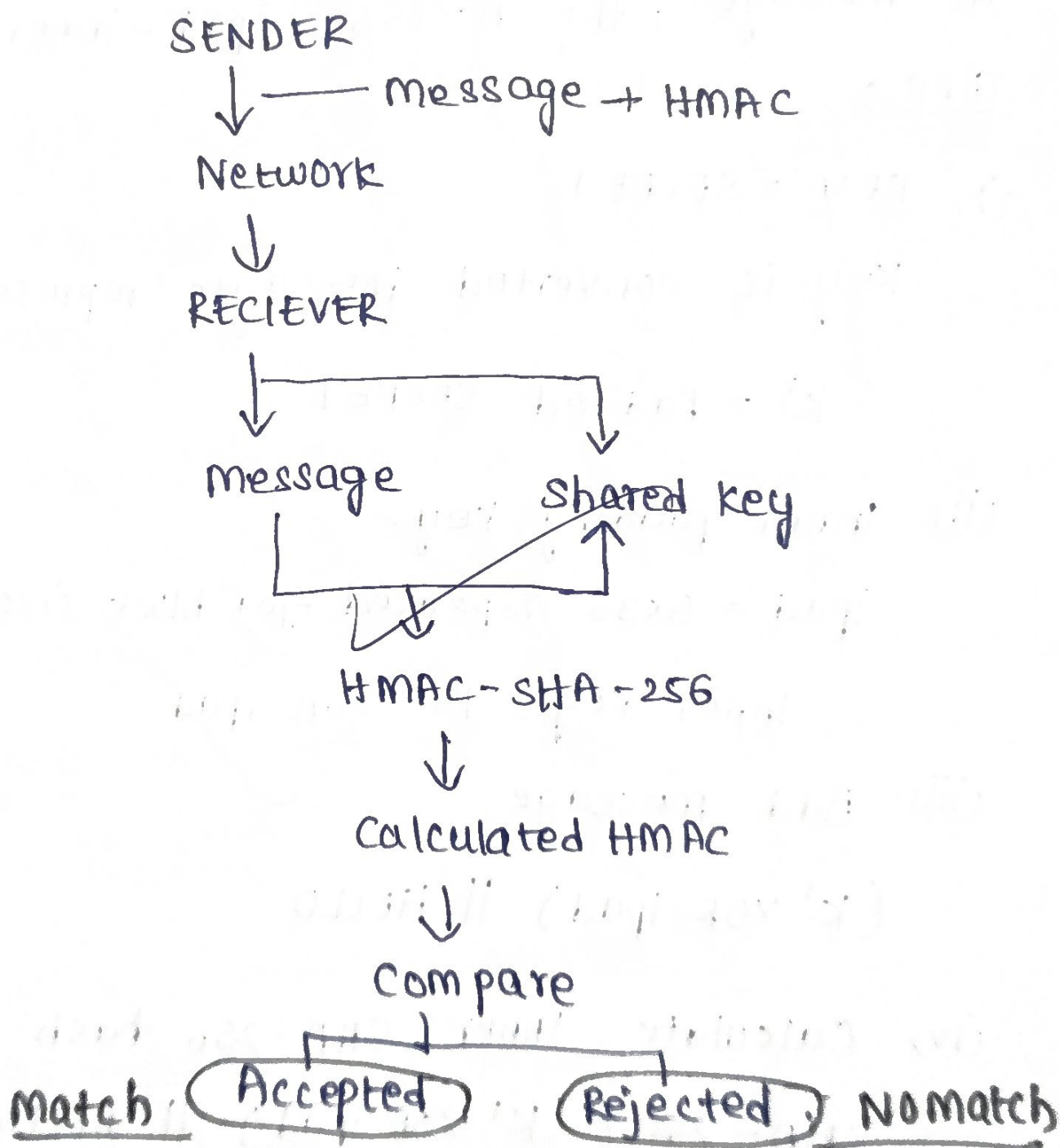
Outer key = $k' \text{ XOR opad}$

(vi) Calculate Final HMAC

$\text{HMAC} = \text{SHA-256}((k' \text{ XOR opad}) \parallel \text{Inner Hash})$

$\therefore$ This is HMAC-SHA-256 Authentication Code.

Verification Flow:-



MD5 (message Digest Algorithm 5) produces a 128-bit hash value. It was widely used in past, but it is now considered cryptographically broken because practical collision attacks has been demonstrated.

• A collision occurs when two different message produce the same hash value.

Message A



MD5



Hash X

Message B



MD5



Hash X

SHA-256 VS MD5		
Feature	MD5	SHA-256
Digest size	128 bits	256 bits
Hexadecimal len	32	64
collision resistance	broken	Stronger
modern security	NO	widely

The weakness is from this

Problem with MD5 :-

- MD5 produces only a 128-bit digest
- Practical collision attacks exist
- It is considered unsuitable for modern cryptography
- It produces smaller security than SHA-256.

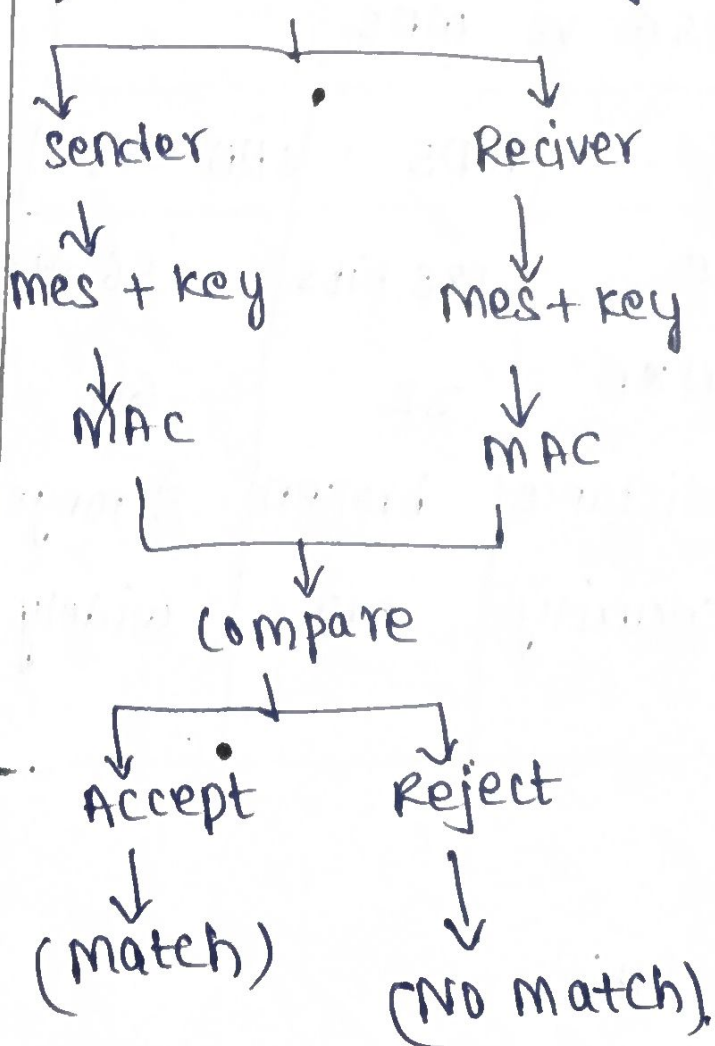
d)

A MAC (Message Authentication Code) and a digital signature are both cryptographic mechanisms that provide message integrity & authentication.

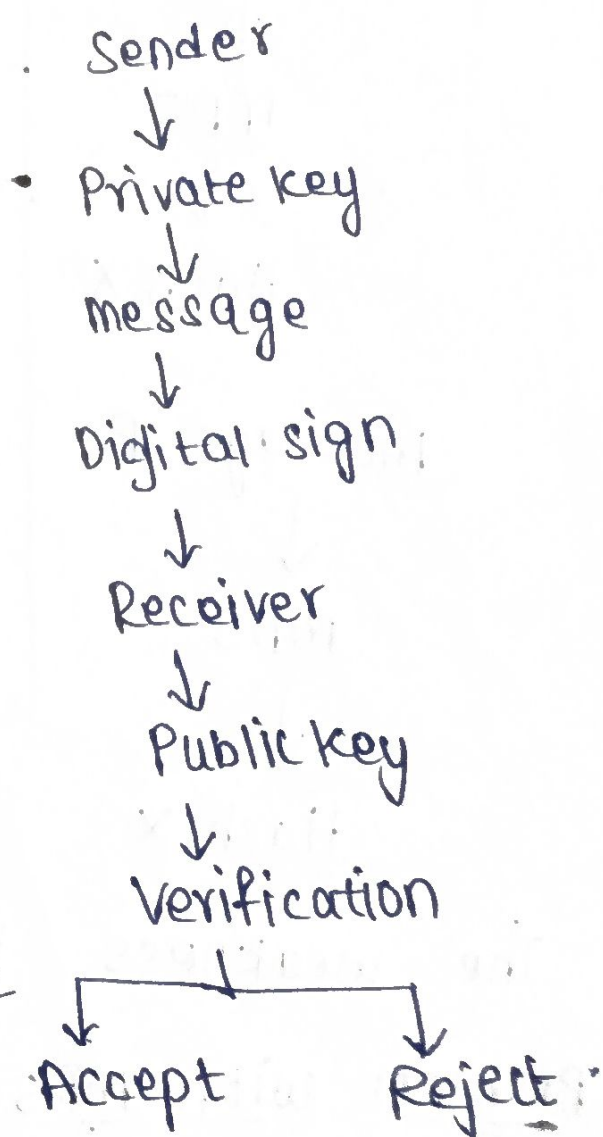
- However, they use different keys & provide different guarantee regarding non-repudiation.

MAC :- Uses shared secret key b/w sender & Receiver

- Shared Secret key



Digital Signature :-



Authentication :-

- * Both MACs and digital signature can authenticate
- * With a MAC, authentication is based on secret key
- * With Digital sign, it is based on control of public key